Topic: The Use of Fixel-Based Analysis in Understanding Early White Matter Degeneration in ALS

Introduction and Background:

ALS and Biomarkers

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease that involves the progressive degeneration of upper and lower motor neurons, which results in progressive muscle weakness and, in some individuals, occurs alongside cognitive impairment. This degeneration is widespread across both grey and white matter, where it is particularly notable within the corticospinal tract (CST) and corpus callosum (CC) (Bede *et al.*, 2015; Zhou *et al.*, 2017; Chen *et al.*, 2021). As the nature of the degeneration varies, reliable and non-invasive *in vivo* exploration of abnormalities in white matter micro and macrostructure in people with ALS (pwALS) is imperative to improving the characterisation of the disease. Much of the evidence of ALS pathology comes from neuropathology studies, which are limited to describing post-mortem cases. Therefore, there is a need for non-invasive quantitative biomarkers (which facilitate the tracking of disease progression as well as the effect on quality of life) that are sensitive to microstructural white matter changes, which is vital especially early in the disease course when progressive pathology is likely subtle (Turner and Verstraete, 2015; Woo *et al.*, 2017).

Diffusion imaging

Diffusion-weighted MRI (DWI) is an imaging method that has been used extensively to characterise cortical microstructure (Martinez-Heras *et al.*, 2021). Conventional radiological imaging methods produce weighted images that highlight the relative differences in signal intensity between tissues. Diffusion-weighted MRI, however, produces diffusion-weighted images (DWI) through tissue's sensitivity to the Brownian motion of water (Bihan and Johansen-Berg, 2012).

In white matter, diffusion is anisotropic (directionally dependent) as axons and myelin restrict water movement perpendicular to the fibre more than movement parallel to it (Beaulieu, 2002). Diffusion tensor imaging (DTI) is a DWI technique that models anisotropy within each voxel as a single tensor, producing quantitative metrics like fractional anisotropy (FA) and mean diffusivity (MD). Histopathological work by Concha et al. (2006) confirmed that reductions in anisotropy can correspond to axonal myelin breakdown, providing biological validation for DTI metrics as markers of microstructural damage. Subsequently, reduced FA in the corticospinal tract has been widely reported as a marker of white matter degeneration in ALS and Multiple Sclerosis (Agosta *et al.*, 2010; Behler *et al.*, 2023; Qin *et al.*, 2024)

Limitations of DTI

The DTI model has significant limitations due to its representation of all intra-voxel diffusion using 3 orthogonal eigenvectors fit it to a single three-dimensional tensor (ellipsoid). This assumes that all diffusion constitutes a single interacting compartment, implying that all tissues within the voxel are congruent and homogeneous (Auriat *et al.*, 2015; Douglas *et al.*, 2015). Therefore, it cannot accurately represent diffusion in fibre bundles that do not fit this assumption, such as crossing or "kissing" fibre populations, which are present in around 90% of white matter voxels (Raffelt *et al.*, 2012). Rather than failing completely at representing the fibres in a voxel, a single tensor cannot disentangle distinct co-existing fibre populations or isolate intra-axonal diffusion from the contributions of extra-axonal water. As a result, its metrics like FA lack fibre-specificity. Additionally, while reductions in anisotropy may indicate impaired white matter integrity, it does not provide information on the underlying physiological cause (Podwalski *et al.*, 2021). This limitation is of particular concern for tracts like the corpus callosum, which are known to cross with other pathways. Another consequence is that degeneration in a secondary fibre population can increase the FA calculated for that voxel, resulting in masked neurodegeneration (Raffelt *et al.*, 2012).

Addressing The Limitations of DTI: FBA

Increased directional sampling across multiple diffusion-weighting strengths (b-values), in particular high b-values, provides the necessary angular and radial data needed to distinguish between separate fibre populations in a single voxel (Raffelt *et al.*, 2017; Jensen and Helpern, 2018). While this high angular sampling strategy was pioneered in single-shell High Angular Resolution Diffusion imaging (HARDI) (Tuch *et al.*, 2002), modern acquisition schemes now offer both increased angular resolution and measurements of diffusion over different weighting strengths. This allows for more advanced modelling of complex tissue microstructure, with fixel-based analysis being one such approach, taking advantage of this depth of information. Instead of averaging the diffusion-weighted signal into one tensor, FBA utilises Constrained Spherical Deconvolution (CSD) to estimate the fibre orientation distribution (FOD) in each voxel (Tournier, Calamante and Connelly, 2007; Jeurissen *et al.*, 2014). The distinct peaks in the FOD correspond with individual fibre populations (fixels), and this allows the metrics of fibre density (FD), fibre cross-section (FC) and their combination (FDC) to be calculated separately for each fibre population (Raffelt *et al.*, 2015, 2017). This fibre-specific approach allows more precise characterisation of white matter degeneration and is applied here to the corticospinal tracts and corpus callosum in ALS.

Previous Applications of FBA:

Previous research has demonstrated the ability of the fibre-specific approach to detect specific information about the type of neurodegeneration, which DTI cannot do. Carandini *et al.*, (2021) applied FBA to brainstem monoaminergic fibre tracts in patients with multiple sclerosis and identified fibre-specific reductions in FD and FC that were not apparent using conventional DTI metrics. The fixel-level changes observed correlated with a specific clinical symptom (fatigue), showing that fibre-specific measures can carry more detailed clinical meaning than what DTI provides.

Additionally, (Ma *et al.*, 2026) recently applied FBA to ALS, analysed diffusion data from the corticospinal tract and corpus callosum and reported fibre-specific reductions in FD and FDC in ALS patients relative to controls, some of which correlated with the degree of motor impairment, as measured with the ALS Functional Rating Scale-Revised (ALSFRS-R). Fibre-specific degeneration is therefore detectable in ALS using FBA, and the CST and CC provide signals that are relevant to the severity of the disease. However, the study relied on basic, single-shell DWI data. So, implementation of a more advanced, multi-shell acquisition protocol should result in a more accurate fibre orientation distribution, and as a result more reliable, fibre-specific FD, FC and FDC measures (Jeurissen *et al.*, 2014; Raffelt *et al.*, 2017).

Aims and Hypothesis:

To address the usage of single-shell data in previous ALS research, this study applies FBA to multi-shell DWI both cross-sectionally and longitudinally to an early-stage ALS cohort. We aim to identify fixel-specific microstructural (FD) and macrostructural (FC, FDC) abnormalities in the CST and CC relative to healthy controls, elucidate the manifestation of the degeneration, and model how this degeneration changes over time. We hypothesise that white matter pathology will worsen in severity over time both relative to controls and in between sessions, with microstructural axonal degradation (indicated by reduced FD) preceding macrostructural tract atrophy (indicated by reduction in FC).

Bibliography:

Agosta, F. *et al.* (2010) 'Assessment of White Matter Tract Damage in Patients with Amyotrophic Lateral Sclerosis: A Diffusion Tensor MR Imaging Tractography Study', *American Journal of Neuroradiology*, 31(8), pp. 1457–1461. Available at: https://doi.org/10.3174/ajnr.A2105.

Auriat, A.M. *et al.* (2015) 'Comparing a diffusion tensor and non-tensor approach to white matter fiber tractography in chronic stroke', *NeuroImage: Clinical*, 7, pp. 771–781. Available at: https://doi.org/10.1016/j.nicl.2015.03.007.

Beaulieu, C. (2002) 'The basis of anisotropic water diffusion in the nervous system – a technical review', *NMR in Biomedicine*, 15(7–8), pp. 435–455. Available at: https://doi.org/10.1002/nbm.782.

Bede, P. *et al.* (2015) 'Patterns of cerebral and cerebellar white matter degeneration in ALS', *Journal of Neurology, Neurosurgery, and Psychiatry*, 86(4), pp. 468–470. Available at: https://doi.org/10.1136/jnnp-2014-308172.

Behler, A. *et al.* (2023) 'Diffusion Tensor Imaging in Amyotrophic Lateral Sclerosis: Machine Learning for Biomarker Development', *International Journal of Molecular Sciences*, 24(3), p. 1911. Available at: https://doi.org/10.3390/ijms24031911.

Bihan, D.L. and Johansen-Berg, H. (2012) 'Diffusion MRI at 25: Exploring brain tissue structure and function', *NeuroImage*, 61(2), pp. 324–341. Available at: https://doi.org/10.1016/j.neuroimage.2011.11.006.

Carandini, T. *et al.* (2021) 'Disruption of brainstem monoaminergic fibre tracts in multiple sclerosis as a putative mechanism for cognitive fatigue: a fixel-based analysis', *NeuroImage : Clinical*, 30, p. 102587. Available at: https://doi.org/10.1016/j.nicl.2021.102587.

Chen, H.-J. *et al.* (2021) 'White matter microstructural impairments in amyotrophic lateral sclerosis: A mean apparent propagator MRI study', *NeuroImage: Clinical*, 32, p. 102863. Available at: https://doi.org/10.1016/j.nicl.2021.102863.

David Raffelt *et al.* (2012) 'Apparent Fibre Density: A novel measure for the analysis of diffusion-weighted magnetic resonance images', *NeuroImage*, 59(4), pp. 3976–3994. Available at: https://doi.org/10.1016/j.neuroimage.2011.10.045.

Douglas, D.B. *et al.* (2015) 'Diffusion Tensor Imaging of TBI: Potentials and Challenges', *Topics in magnetic resonance imaging : TMRI*, 24(5), pp. 241–251. Available at: https://doi.org/10.1097/RMR.0000000000000062.

Jensen, J.H. and Helpern, J.A. (2018) 'Characterizing intra-axonal water diffusion with direction-averaged triple diffusion encoding MRI', *NMR in biomedicine*, 31(7), p. e3930. Available at: https://doi.org/10.1002/nbm.3930.

Jeurissen, B. *et al.* (2014) 'Multi-tissue constrained spherical deconvolution for improved analysis of multi-shell diffusion MRI data', *NeuroImage*, 103, pp. 411–426. Available at: https://doi.org/10.1016/j.neuroimage.2014.07.061.

Ma, M. *et al.* (2026) 'Fixel-based analysis to detect early fiber-specific white matter alterations in sporadic patients with amyotrophic lateral sclerosis', *Brain Imaging and Behavior*, 20(1), p. 16. Available at: https://doi.org/10.1007/s11682-026-01097-y.

Martinez-Heras, E. *et al.* (2021) ‘Diffusion-Weighted Imaging: Recent Advances and Applications’, *Seminars in Ultrasound, CT and MRI*, 42(5), pp. 490–506. Available at: https://doi.org/10.1053/j.sult.2021.07.006.

Podwalski, P. *et al.* (2021) ‘Magnetic resonance diffusion tensor imaging in psychiatry: a narrative review of its potential role in diagnosis’, *Pharmacological Reports*, 73(1), pp. 43–56. Available at: https://doi.org/10.1007/s43440-020-00177-0.

Qin, J. *et al.* (2024) ‘Identifying amyotrophic lateral sclerosis using diffusion tensor imaging, and correlation with neurofilament markers’, *Scientific Reports*, 14, p. 28110. Available at: https://doi.org/10.1038/s41598-024-79511-y.

Raffelt, D.A. *et al.* (2015) ‘Connectivity-based fixel enhancement: Whole-brain statistical analysis of diffusion MRI measures in the presence of crossing fibres’, *Neuroimage*, 117, pp. 40–55. Available at: https://doi.org/10.1016/j.neuroimage.2015.05.039.

Raffelt, D.A. *et al.* (2017) ‘Investigating white matter fibre density and morphology using fixel-based analysis’, *NeuroImage*, 144(Pt A), pp. 58–73. Available at: https://doi.org/10.1016/j.neuroimage.2016.09.029.

Tournier, J.-D., Calamante, F. and Connelly, A. (2007) ‘Robust determination of the fibre orientation distribution in diffusion MRI: Non-negativity constrained super-resolved spherical deconvolution’, *NeuroImage*, 35(4), pp. 1459–1472. Available at: https://doi.org/10.1016/j.neuroimage.2007.02.016.

Tuch, D.S. *et al.* (2002) ‘High angular resolution diffusion imaging reveals intravoxel white matter fiber heterogeneity’, *Magnetic Resonance in Medicine*, 48(4), pp. 577–582. Available at: https://doi.org/10.1002/mrm.10268.

Turner, M.R. and Verstraete, E. (2015) ‘What Does Imaging Reveal About the Pathology of Amyotrophic Lateral Sclerosis?’, *Current Neurology and Neuroscience Reports*, 15(7), p. 45. Available at: https://doi.org/10.1007/s11910-015-0569-6.

Woo, C.-W. *et al.* (2017) ‘Building better biomarkers: brain models in translational neuroimaging’, *Nature neuroscience*, 20(3), pp. 365–377. Available at: https://doi.org/10.1038/nn.4478.

Zhou, T. *et al.* (2017) ‘Implications of white matter damage in amyotrophic lateral sclerosis’, *Molecular Medicine Reports*, 16(4), pp. 4379–4392. Available at: https://doi.org/10.3892/mmr.2017.7186.